

SEDAR 69 ASFMC Atlantic Menhaden

Assessment Process

Benchmark Assessment Research Recommendations

Short Term

1. Continue current level of sampling from bait fisheries, particularly in the Mid-Atlantic and New England. Analyze sampling adequacy of the reduction fishery and effectively sample areas outside of that fishery (e.g., work with industry and states to collect age structure data and biological data outside the range of the fishery).
2. Place observers on boats to collect at-sea samples from purse-seine sets, or collect samples at dockside during vessel pump-out operations (as opposed to current top of hold sampling) to address sampling adequacy.
3. Evaluate which proportion of bait landings by state are captured by gear versus which proportion are sampled for length and age composition to determine if current biosampling requirements are appropriate and adequate.
4. Continue to improve data validation processes for the bait fishery through ACCSP.
5. Conduct an ageing workshop to assess precision and error among readers with the intention of switching bait fishery age reading to state ageing labs.
6. Re-age historic old age samples (i.e., ages >7) to confirm the max age of Atlantic menhaden.
7. Investigate the relationship between fish size and school size to address selectivity (specifically addressing fisher behavior related to harvest of specific school sizes).
8. Investigate the relationship between fish size and distance from shore (addressing selectivity).

Long Term

1. Develop and implement a menhaden-specific, multi-year coastwide fishery-independent index of adult abundance-at-age with ground-truthing for biological information (e.g., size and age composition). A sound statistical design is essential. Ideally, it should be done coast-wide, but area-specific surveys that cover the majority of the population and are more cost-effective could provide substantial improvements over the indices currently used in the assessment.
2. Continue age-specific studies on spatial and temporal dynamics of spawning (where, how often, how much of the year, batch spawning, etc.)
3. Conduct an ageing validation study, making sure to sample older age classes.
4. Continue to investigate environmental covariates related to productivity and recruitment on a temporal and spatial scale.

5. Consider other ageing methods for the future, such as the use of Fourier transform near infrared spectroscopy (FT-NIRS).

Assessment Methods

Short Term

1. Investigate index standardization to improve CVs and explore methods of combining indices at a regional or coastwide level.
2. Explore the covariance between life history parameters to improve the understanding of uncertainty in the model.
3. Explore the error structure between MCMC and MCB.
4. Perform simulation testing on the Deyle et al. method used in the projections and determine if recruitment is accurately tracked by the method and improve short term projections.
5. Conduct a Management Strategy Evaluation (MSE).

Long Term

1. Continue to monitor model diagnostics given that the model is not robust to anomalous year-classes in the terminal year.
2. Develop a seasonal spatially-explicit model once sufficient age-specific data on movement rates of menhaden are available.

Ecological Reference Points Research recommendations

The ERP WG endorsed the research recommendations laid out in the single-species assessment to improve the understanding of Atlantic menhaden population dynamics, especially the recommendations to develop an Atlantic menhaden-specific coastwide fishery-independent index of adult abundance and to continue to investigate environmental covariates related to productivity and recruitment on a temporal and spatial scale.

In addition, the ERP WG identified a number of research needs to improve the multispecies modeling efforts and the development of ecological reference points for Atlantic menhaden, as well as process considerations to fully implement ecosystem-based fishery management.

Future Research and Data Collection

Short term

1. Expand collection of diet and condition data along the Atlantic coast to provide seasonally and regionally stratified annual, year-round monitoring of key predator diets to provide information on prey abundance and predator consumption. This could be done through existing data collection programs.

Long term

1. Improve monitoring of population trends and diet data in non-fish predators (e.g., birds, marine mammals) and data-poor prey species (e.g., bay anchovies, sand eels, benthic invertebrates, zooplankton, and phytoplankton) to better characterize the importance of Atlantic menhaden and other forage species to the ecosystem dynamics.

Modeling Needs*Short term*

1. Conduct a management-strategy evaluation (MSE) to identify harvest strategies that will maximize the likelihood of achieving the identified ecosystem management objectives.
2. Continue development of the NWACS-MICE model to incorporate recruitment deviations (from external models or primary productivity time series) to better capture the productivity dynamics of Atlantic menhaden and other species.
3. Continue development of the VADER model to include bottom-up effects of Atlantic menhaden abundance on key predator species.
4. Continue development of the NWACS-FULL model to bring other species up to date and continue exploring the impacts of fishing on higher trophic level predators like birds and mammals.

Review Workshop**Benchmark Review Research Recommendations**

A number of short and longer term recommendations were made for research, data collection, modelling and management by the TC. The panel was in general agreement with these recommendations. The panel offered the following supplementary comments.

In the short term, the panel strongly encourages a thorough exploration of options for adding a survey or fishery-dependent dataset that more representatively samples the larger, older fish in the population. One option might be expansion of sampling from the bait fishery. The panel also suggested further exploration of the size frequency data from the existing fishery independent surveys to assess the internal consistency of the surveys (i.e., cohort tracking) as a further screening tool for inclusion/deletion of surveys in future assessments (see ToR 1). Collection of age data for the existing fishery independent surveys is also considered a priority, and has been recommended in the past (SEDAR 40 - 2015). The panel also supported the TC's recommendations for the development of a coast-wide fishery-independent index of abundance-at-age given that none of the existing fishery-independent surveys are specifically directed towards menhaden.

One of the recommendations under assessment methods was to conduct a Management Strategy Evaluation (MSE). The panel agrees that an MSE would be valuable for evaluating the robustness to uncertainty of reference points and control rules informed by the single species assessment. Another potential benefit would be to evaluate the benefits of improved data collection and biological sampling.

However, there are “devils in the details”: decisions about how to structure the MSE will require careful thought to avoid progress on management being impeded by a process that could take several years and require a large commitment of resources. Finally, if an MSE is to be undertaken, the panel recommends that the effort be framed in the context of Ecosystem Reference Points rather than single species management, as discussed in the review of the ERP report.

With respect to research recommendations regarding assessment methods, the panel noted that while the automatic differentiation optimizer in ADMB is fast and therefore efficient, it is vulnerable to false convergence problems. It is therefore important the performance of the optimization be examined carefully before presenting model solutions. There were instances with the single species BAM model (see ToR 2) where diagnostics suggested an optimization failure. The panel therefore strongly recommends that a so-called jitter analysis be performed on any model solution. This involves running a large number of optimizations, each one with slightly different starting parameters, to increase confidence that the final solution represents solution global optimum.

Ecological Reference Points Review Research Recommendations

The report included a number of recommendations for future research, data collection, modelling and management, for both the short and long term. These included expanding collection of diet and condition data, to include non-fish predators and data-poor prey species, to conduct management-strategy evaluation (MSE) to identify harvest strategies that will meet ecosystem management objectives, and to continue the development of the NWACS-MICE, NWACS-FULL and VADER models.

The panel fully supported these recommendations. A number of additional specific recommendations for research on the ERP models and assessment methods to inform these models are presented earlier in this report in the context of other ToRs.

We recognized the potential strategic importance of conducting an MSE and noted the benefit of having already completed an Ecosystem Management Objectives Workshop to identify fundamental ecosystem management objectives for Atlantic menhaden. An MSE could be used to examine alternative scenarios to ensure the management advice is robust and to fully explore tradeoffs among alternative management strategies. However, as noted under ToR 8 for the single species review, it will be important to plan an MSE process carefully, to avoid progress on management being impeded by a process that could take several years and require a large commitment of resources.